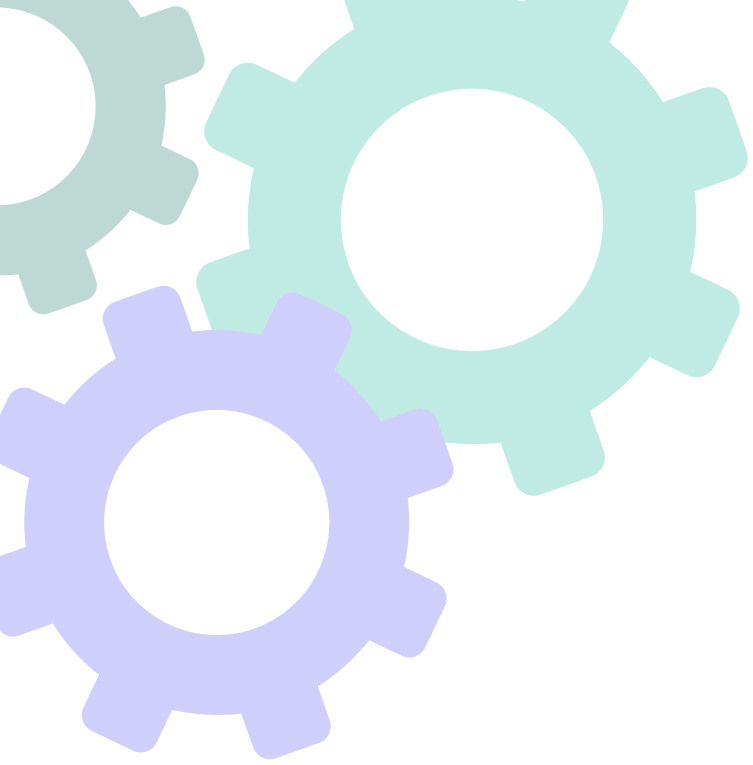
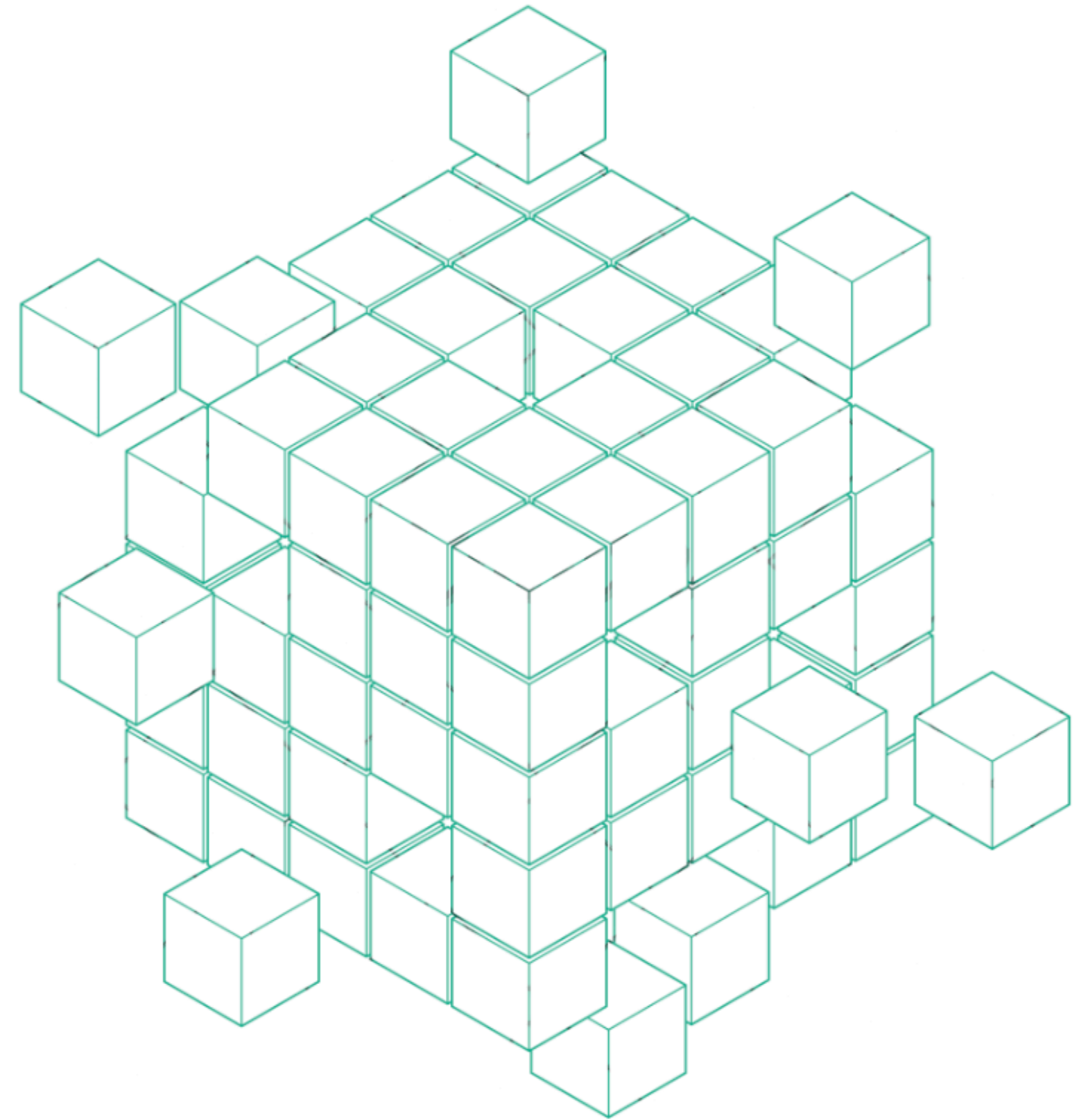




Loan Risk Predictor ML System Design

A governed path from application data to scalable champion model

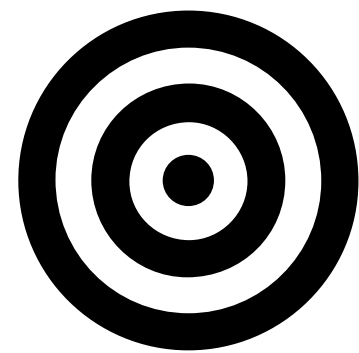
Tan Wei Yang (Ventus)



LightGBM Machine Learning Model Definition

The model predicts adverse repayment before pricing

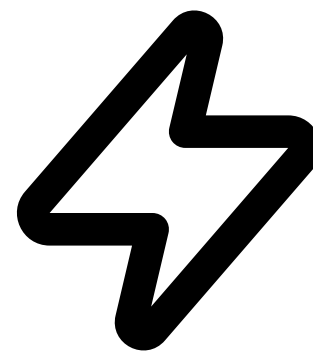
TARGET



How risky is a loan

Risk is defined by the probability of an adverse repayment outcome

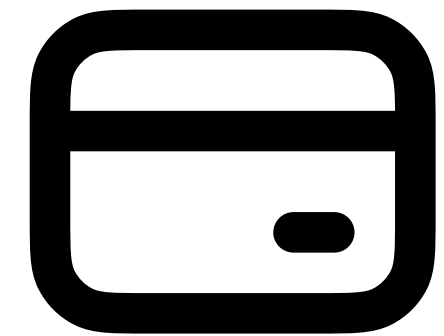
GOOD OUTCOME



A funded loan is fully paid off

Paid Off Loan

BAD OUTCOME



Collection, bankruptcy, or charge-off

Internal Collection
External Collection
Settled Bankruptcy
Charged Off

Feature Contract & Leakage Control

Only information available before pricing enters the model; the same ordered schema runs in training and inference.



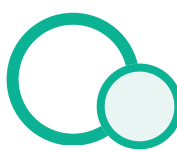
PREDICTION POINT

Known when the application is scored

- Loan amount
- Lead type and state
- Application month, day and hour
- Clarity report signals
- Prior-loan history

TRAINING–SERVING PARITY

The same feature definitions, data types, transformations and missing-value rules are applied in training and inference.



POST-DECISION & OUTCOME DATA

Excluded because it is created after, or influenced by the decision

- APR and scheduled payment
- Approval / origination outputs
- Final loan status
- Repayment records
- Raw identifiers (i.e. Loan ID)

TIME-AWARE EVALUATION

Train, validation and test are split chronologically so later cohorts stay genuinely unseen.

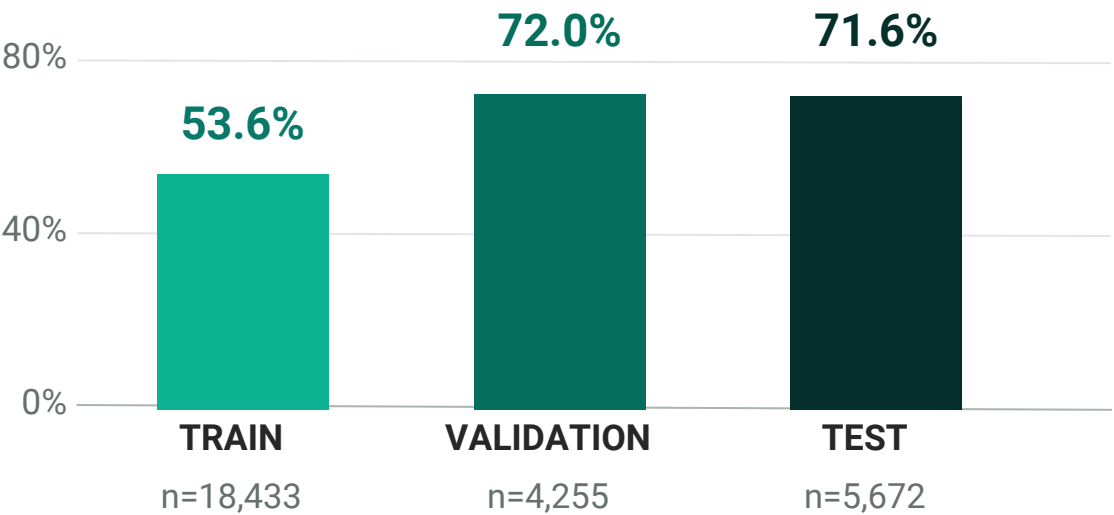
No future-to-past leakage

INITIAL OUTCOME SPLIT

28,360 resolved funded loans



CHRONOLOGICAL ADVERSE RATE



Later cohorts have a higher adverse rate.

Train on earlier loans and test on later loans so evaluation reflects this change over time

Model Metrics Evaluation

Ranking metrics establish model quality before selecting a business operating threshold.

ROC-AUC

THRESHOLD-INDEPENDENT DISCRIMINATION

Tests whether adverse loans receive higher risk scores than paid-off loans across every possible threshold.

PR-AUC

ADVERSE-CLASS PRECISION & RECALL

Shows how well the model finds adverse loans without incorrectly flagging loans that will be paid off.

CHRONOLOGICAL TEST-SET BENCHMARK

MODEL	ROC-AUC	PR-AUC	Log loss
Constant predictor random ranking / prevalence baseline	0.500	0.716	0.664
Logistic regression linear benchmark	0.724	0.837	0.562
LightGBM selected ranking model	0.737	0.849	0.615

RANKING GAIN

+0.237

ROC-AUC vs random

+0.133

PR-AUC vs 71.6% prevalence

LightGBM improves ranking over both baselines on the untouched test cohort.

THRESHOLD

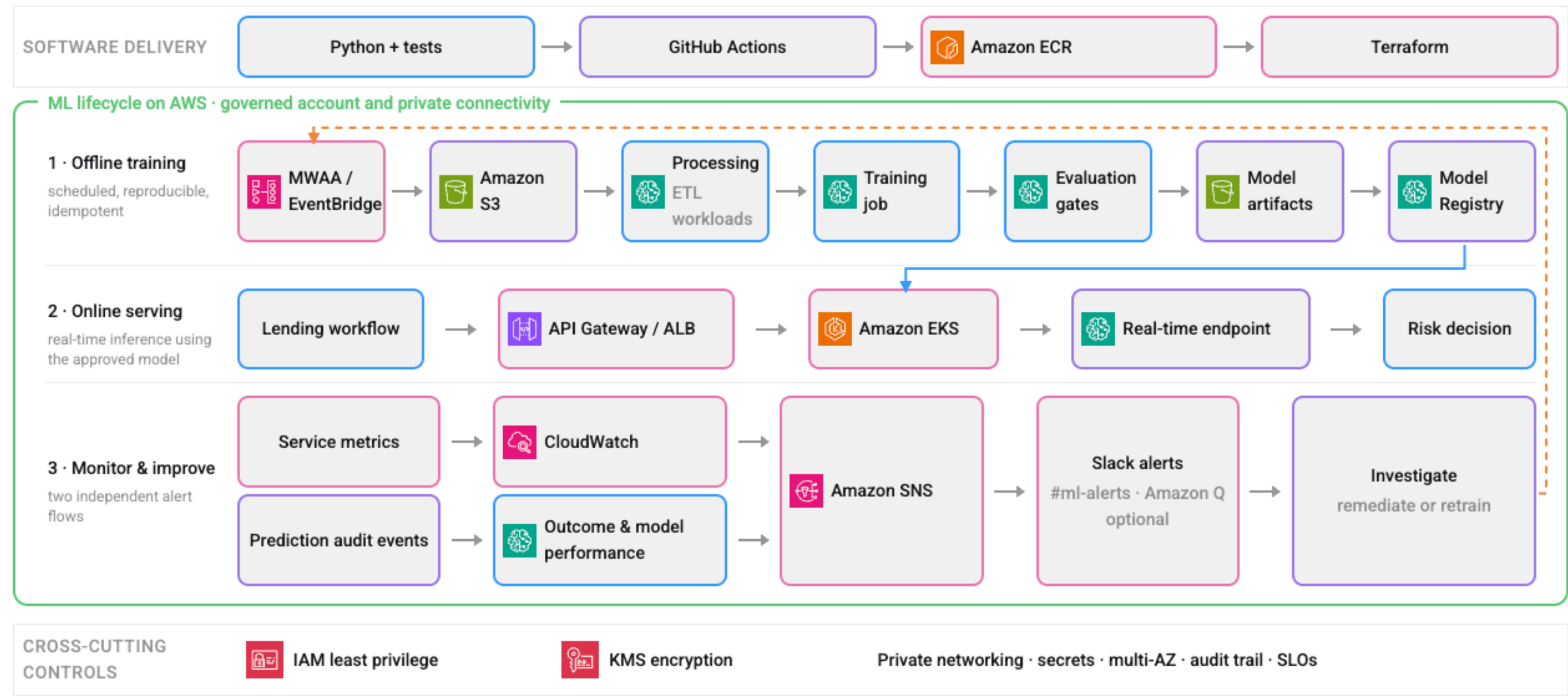
Reduce false negatives while accepting a controlled increase in false positives; Optimal threshold would still need to be discussed with business.

RANKING ≠ CALIBRATION

Assess probability calibration separately before pricing.

Production Target Architecture

Highlight: ● All ● ML engineering ● MLOps platform ● Shared boundary



The local POC implements the same boundaries using Docker Compose and a filesystem-backed registry.

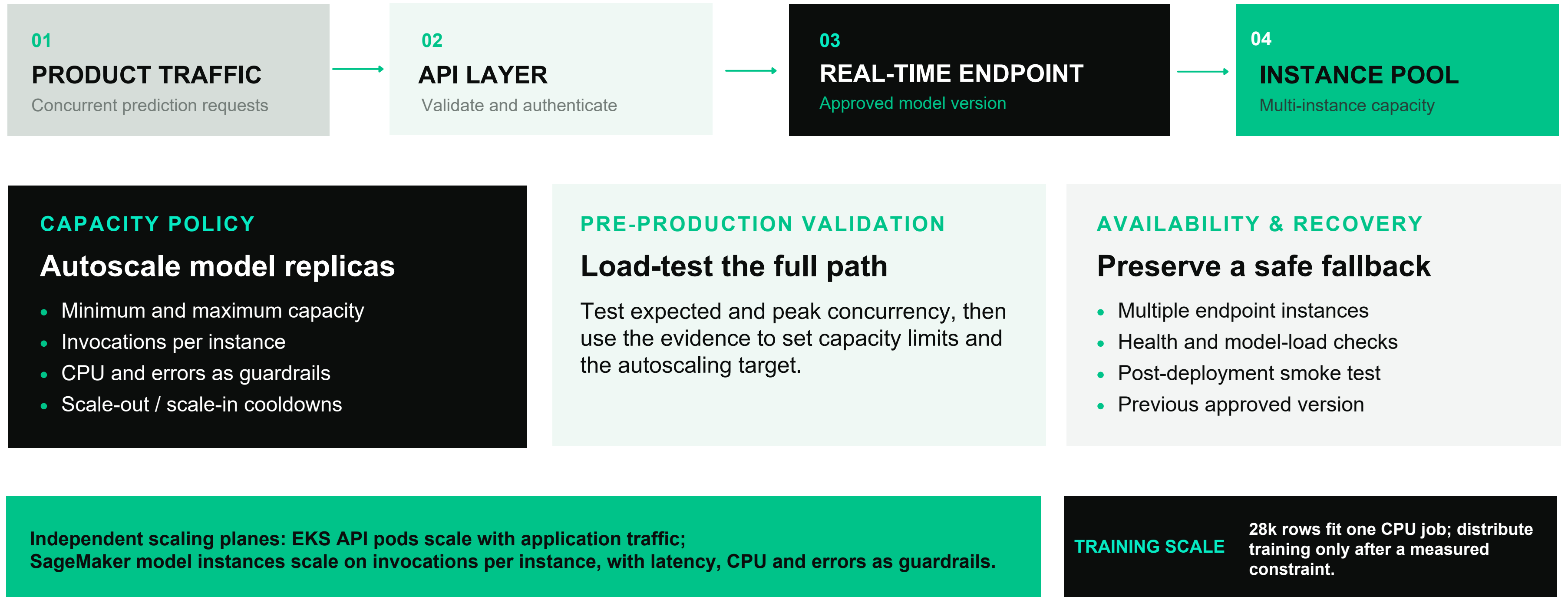
Model Training & Promotion Controls

EVIDENCE	PROMOTION	TEST
Gate the candidate Check data quality, minimum sample size, ROC-AUC, and log loss. Apply predictive-quality acceptance thresholds.	Approve and deploy Register an immutable version with evaluation evidence. Keeps previous version available.	Smoke-test or restore Sends representative request to new endpoint. Restore the previous approved version if validations fail.
<pre>minimum_test_rows: 1000 minimum_roc_auc: 0.65 maximum_log_loss: 0.75 roc_auc_regression_tolerance: 0.01</pre> <i>promotion.yaml example</i>	<pre>{ "actor": "admin@example.com", "previous_version": "20260806T084258Z-b8f76871", "reason": "testing_rollback", "updated_at_utc": "2026-08-07T01:39:55.837323+00:00", "version": "20260807T013638Z-8a8adcfe" }</pre> <i>champion.json example</i>	<pre>{ "applicationDate": "2020-01-15T12:00:00Z", "has_clarity_report": false, "leadCost": 0, "leadType": "bvMandatory", "loanAmount": 0, "payFrequency": "B", "state": "AK" }</pre> <i>smoke_request.json example</i>

CONTROL MAPPING	LOCAL POC	promotion.yaml	→	Immutable local bundle	→	champion.json	→	FastAPI smoke test
	PRODUCTION	Pipeline conditions	→	SageMaker Model Registry	→	Endpoint configuration	→	Smoke test & rollback

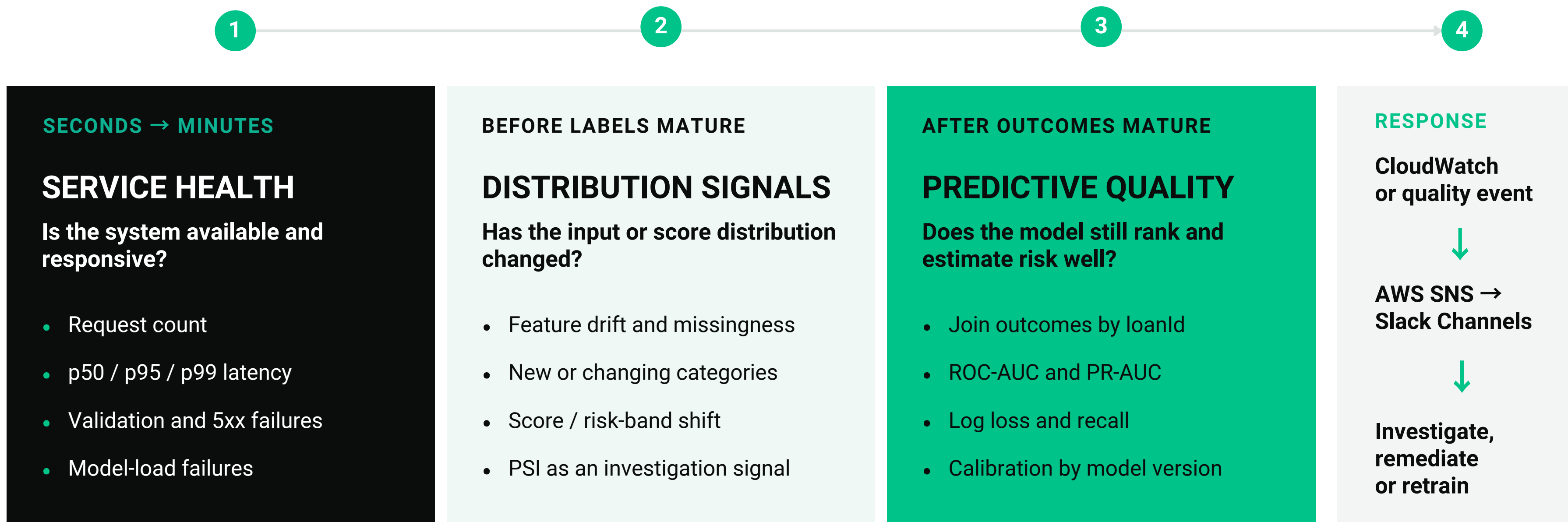
Model Serving & Autoscaling Strategy

A managed real-time endpoint provides one stable inference interface while scaling model replicas behind it.



Service Health & Model Quality Monitoring

Each evidence horizon answers a different question and triggers a controlled response.



AUDIT CONTEXT

Every event carries the model version and feature-contract version.

Alerts open investigation or controlled retraining; only evidence gates and approval can change production.

Design Summary

A controlled lifecycle connecting model development, safe delivery and continuous oversight.

POC demonstrates reproducible training, controlled promotion, traceable serving and tested recovery.



MODEL INTEGRITY

Predictions use only information available at decision time

- Pre-decision scoring point
- Leakage-controlled labels and features
- ROC-AUC, PR-AUC and log-loss evaluation



SAFE DELIVERY

Only validated, versioned models become the champion

- Reproducible, versioned artifacts
- Evidence-based promotion gates
- Controlled champion change with rollback



CONTINUOUS OVERSIGHT

Monitoring covers both immediate health and delayed quality check

- Live latency, errors and volume
- Feature and score drift signals
- Outcome-based performance by version

Every prediction is traceable, every promotion is evidence-gated, and every release is reversible.

POC Demo: Scoring & Model Operations

LRC

LoanRiskCalculator

Decision intelligence

Scoring service online

Admin login

PRE-PRICING RISK ASSESSMENT

Make the next lending decision with clarity.

Estimate adverse repayment risk using the currently promoted model, with a traceable version and decision on every request.

RUNTIME METRICS

Requests 0 Failures 0 Latency 0.00s

Risk bands low 0 / medium 0 / high 0

Single Batch

01

Application details

Enter information available before pricing.

Application date

07/08/2026

Loan amount

\$ 500

Lead acquisition cost

\$ 25

Lead type

Bank verification mandatory

Pay frequency

Biweekly

State

California

☐ Clarity report available

Include the presence signal in this assessment.

Run risk assessment

→

02

Decision result

Model score and operational action.

Ready when you are

Complete the application details to generate a versioned risk assessment.

✓ Probability and risk band

✓ Pass or review decision

✓ Traceable request and model version

Risk scoring workspace

LRC

LoanRiskCalculator

Decision intelligence

Scoring workspace

Sign out

RESTRICTED ACCESS

Model operations

Signed in as admin@example.com. Use only during approved model governance or an investigated incident. Every change is recorded with your identity and reason.

Reason

e.g. Approved rollback after drift investigation

20260807T024432Z-9cb43c13

Active champion

07/08/2026, 03:44:33

roc_auc: 0.737

pr_auc: 0.849

log_loss: 0.605

20260807T024411Z-52b5f7f2

Rollback target

07/08/2026, 03:44:13

roc_auc: 0.736

pr_auc: 0.848

log_loss: 0.615

Promote

20260807T024231Z-671da447

accepted

07/08/2026, 03:42:35

roc_auc: 0.737

pr_auc: 0.849

log_loss: 0.615

Promote

Roll back to previous champion

Enter a reason of at least five characters before changing the champion.

Controlled promotion and rollback

10

Thank You!
Questions & Discussions

Slides are for MoneyLion Senior ML Engineer 2026 Assessment only